Reading structured data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Reading structured data before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Dictionaries and records

Without opening the old lesson, explain one key idea from Dictionaries and records. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Functions and arguments

Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Reading structured data trace and record the first step you can explain without help.

Use W2: Trace all updates for Reading structured data. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Reading structured data explanation against the supplied facts before discussing it.

Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Preserve records and report measured coverage

> Trace the named **field**s and missing-value policy. Predict the **output**, state the observed total and mean denominator, and report coverage. What changes if all three rows are empty records?

```
rows = [{"score": 2}, {"score": 6}, {}]
values = []
for row in rows:
    value = row.get("score")
    if value is not None:
        values.append(value)
observed_total = sum(values)
if len(values) > 0:
    mean = observed_total / len(values)
else:
    mean = None
print(observed_total, len(values), len(rows), mean)
```

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Read a row

> Read a row. Apply the supplied Reading structured data facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Reading structured data", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

In the worked **dataset**, state the number of records and the spare spaces in the second record.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace all updates

Trace all updates. Apply the supplied Reading structured data facts and keep your reasoning.

For spare contributions 2, 4 and 1, trace total starting at 0.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Repair initialization

> Repair initialization. Apply the supplied Reading structured data facts and keep your reasoning.

A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

My reasoning and evidence

W4 Process new records independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Process new records independently. Apply the supplied Reading structured data facts and keep your reasoning.

Write a loop for records (capacity 9, used 2), (capacity 5, used 1), and (capacity 4, used 4). Predict the total. Reverse the order and test an empty list.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Interpret an empty result

Interpret an empty result. Apply the supplied Reading structured data facts and keep your reasoning.

An empty **dataset** produces total 0. Can you conclude that every cupboard is full?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

Connect Reading structured data to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled reading structured data **evidence** sheet to Python readiness portfolio. Show how you can **process** a list of records consistently.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
>
> _____
>
> _____
>
> _____
>
> _____

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

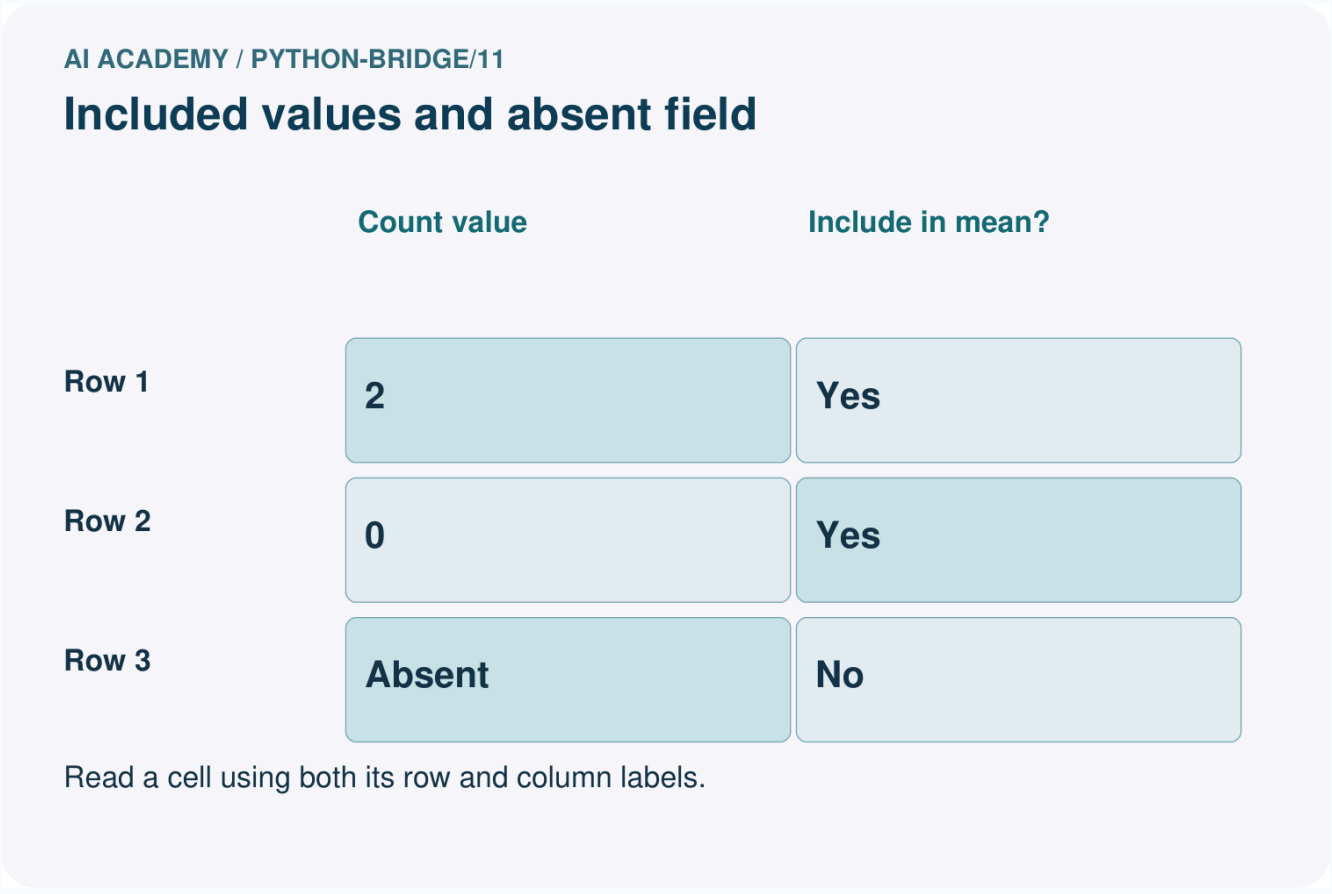
Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Reading structured data

> Read rows with a missing-field policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count. Explain the labelled step before checking the changed condition.

Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

Python 3 records: [{"count":2},{"count":0},{}]. Policy: include existing numeric counts, including zero; skip an absent **field**.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/11

Reading structured data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect three rows

Two rows contain count; the third does not.



2. Accumulate values

The included values sum to 2+0=2, with denominator 2.



3. Report coverage and mean

Mean is 1; measured coverage is 2 of 3 rows.

CHANGE ONE THING

Add {"count":4} as a fourth row.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Mean is 1; measured coverage is 2 of 3 rows.

1. Inspect three rows

2. Report coverage and mean

3. Accumulate values

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report coverage and mean Mean is 1; measured coverage is 2 of 3 rows. Structured data need a row policy before aggregation.

Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add {"count":4} as a fourth row. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4. Only the added recorded value contributes; the existing absent **field** remains skipped.

Repair a missing building block

Review Dictionaries and records (python-bridge/10). Goal: Read and update named dictionary **field**s. Short practice: Predict all three original **output**s.

Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.. Return to this week and explain what you changed.

Review Functions and arguments (python-bridge/8). Goal: Distinguish a function definition from a function call. Short practice: Match every argument to a parameter for the two calls.

Why do **output**s differ, and what prints when both calls are removed?. Return to this week and explain what you changed.

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

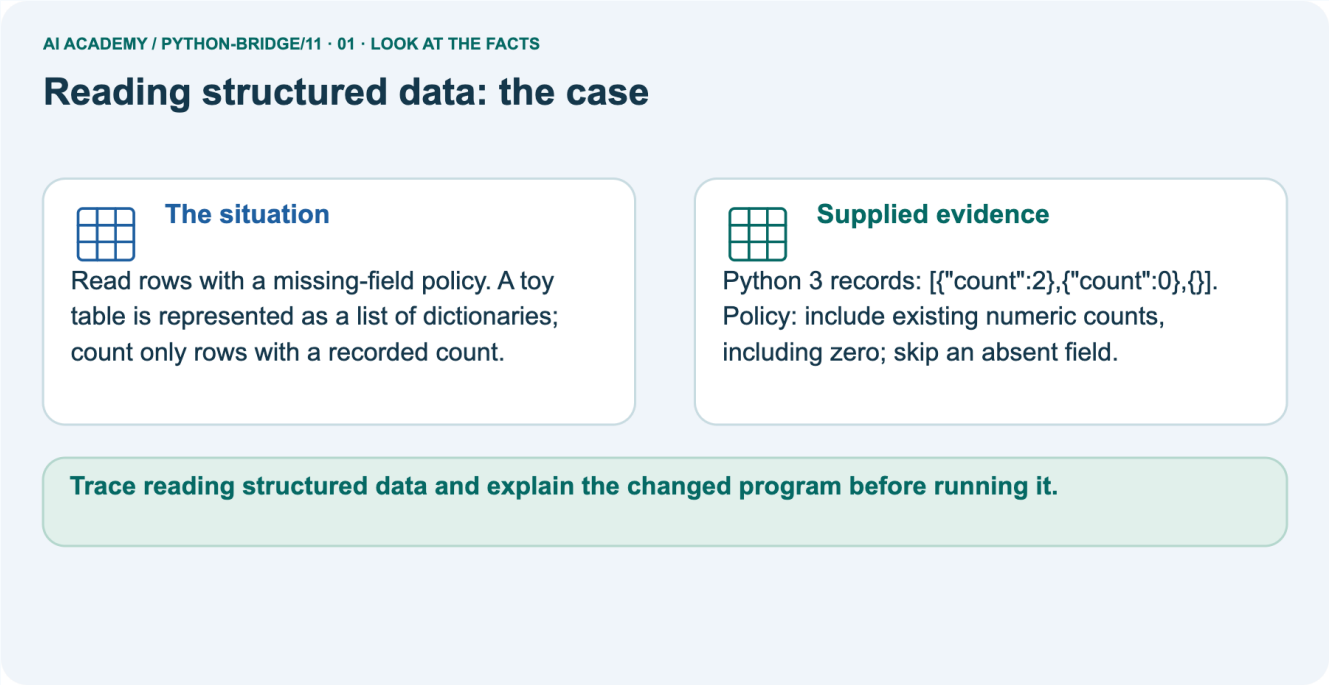
> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Reading structured data picture from its labels, then record a relevant limit.

01. Reading structured data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

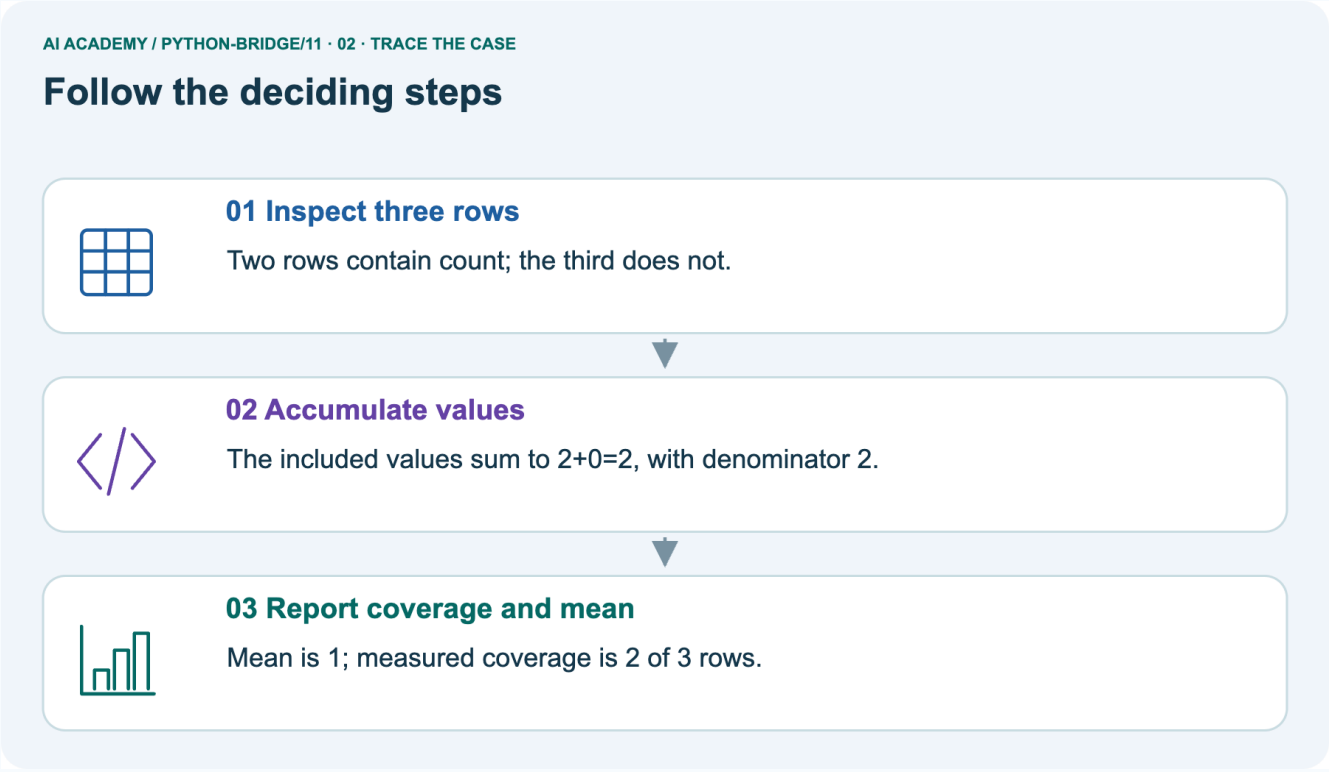
Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count. Supplied facts: Python 3 records: [{"count":2},{"count":0},{}].

Policy: include existing numeric counts, including zero; skip an absent **field**.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



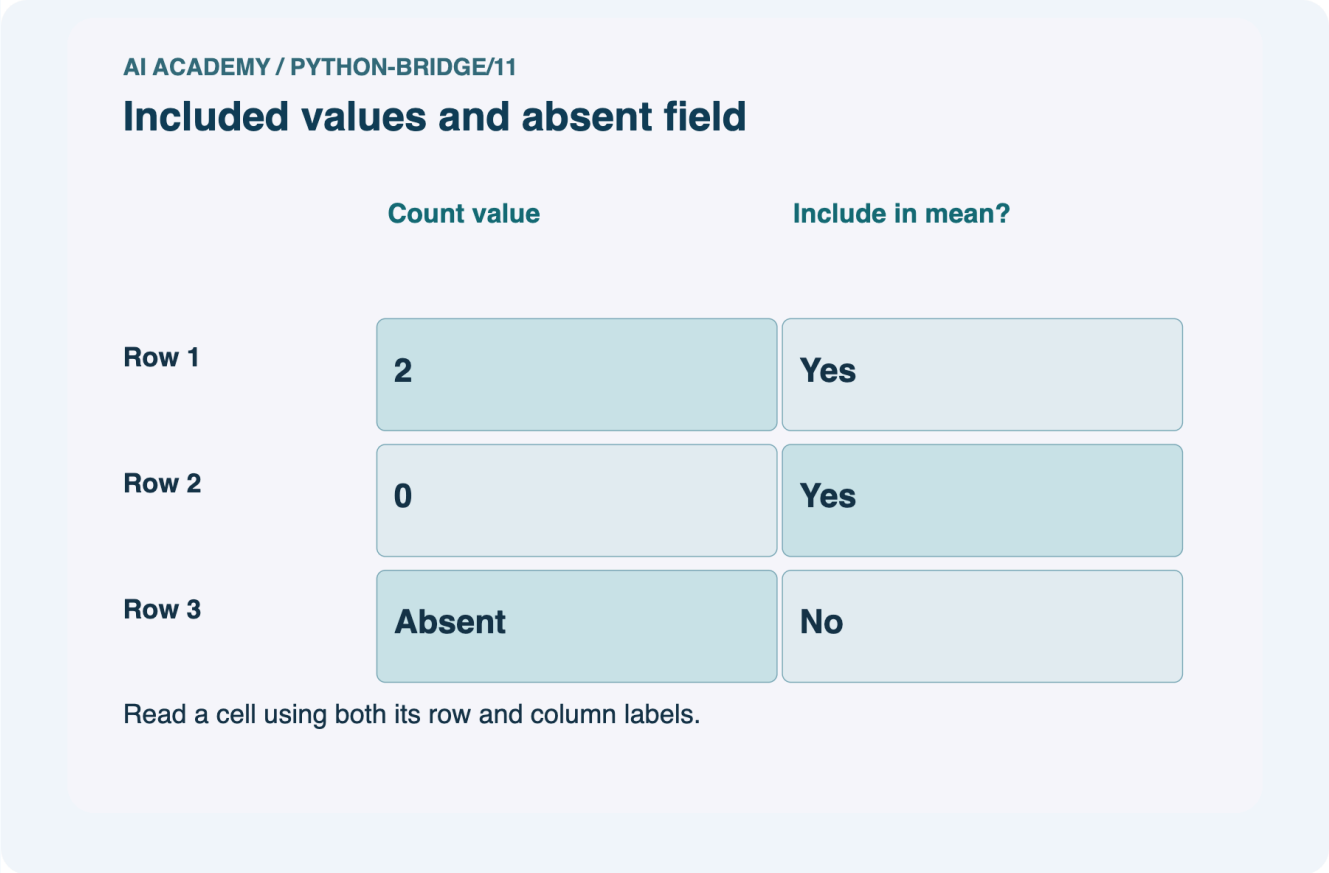
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect three rows: Two rows contain count; the third does not. → Accumulate values: The included values sum to 2+0=2, with denominator 2. → Report coverage and mean: Mean is 1; measured coverage is 2 of 3 rows.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Included values and absent field



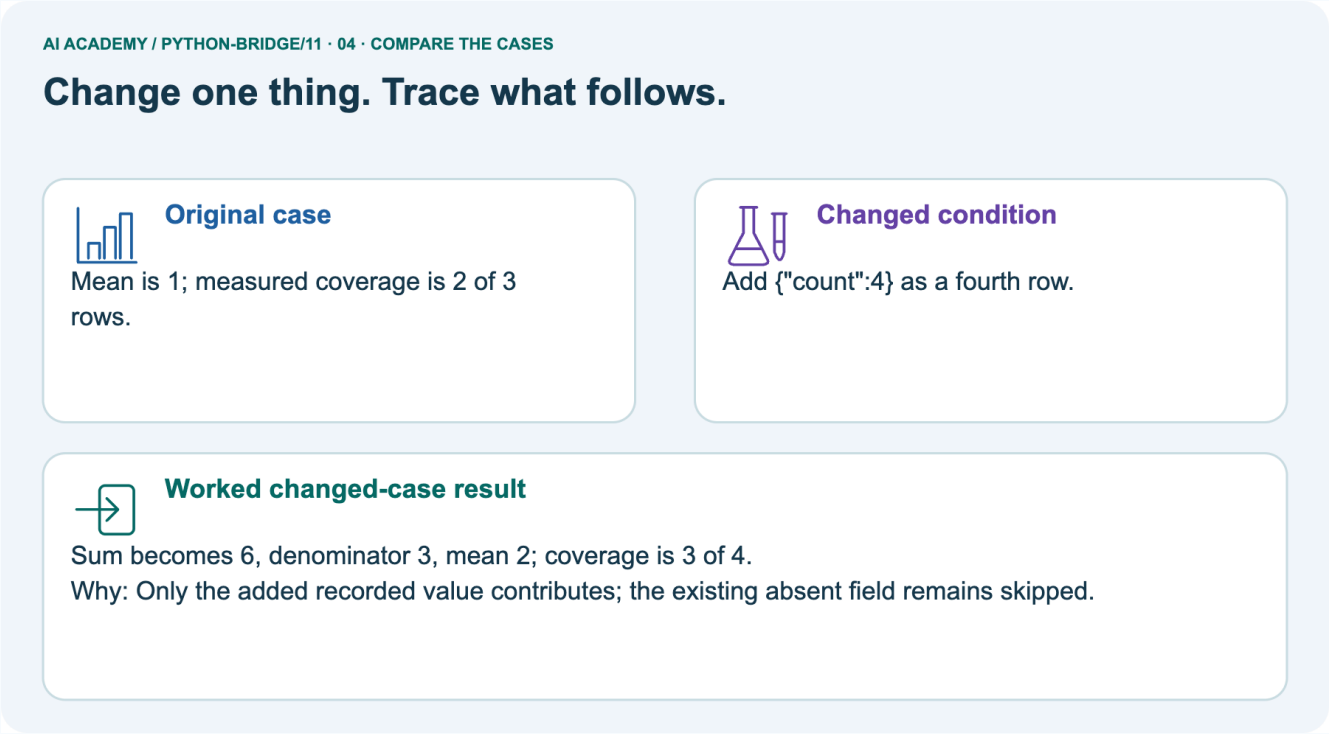
Follow the labelled picture with the matching explanation. Open the vector diagram

Included values and absent **field**. Row 1: Count value = 2, Include in mean? = Yes. Row 2: Count value = 0, Include in mean? = Yes. Row 3: Count value = Absent, Include in mean? = No.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

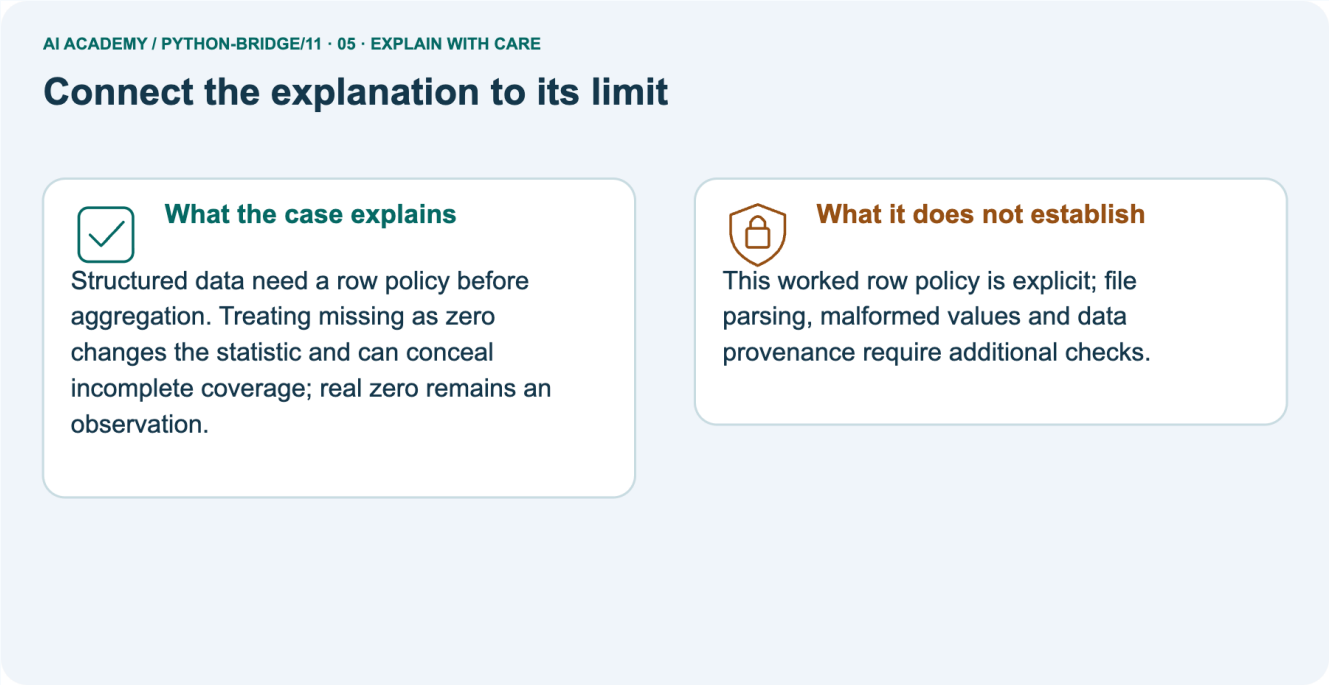
Original case: Mean is 1; measured coverage is 2 of 3 rows. Changed condition: Add {"count":4} as a fourth row. Expected result: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

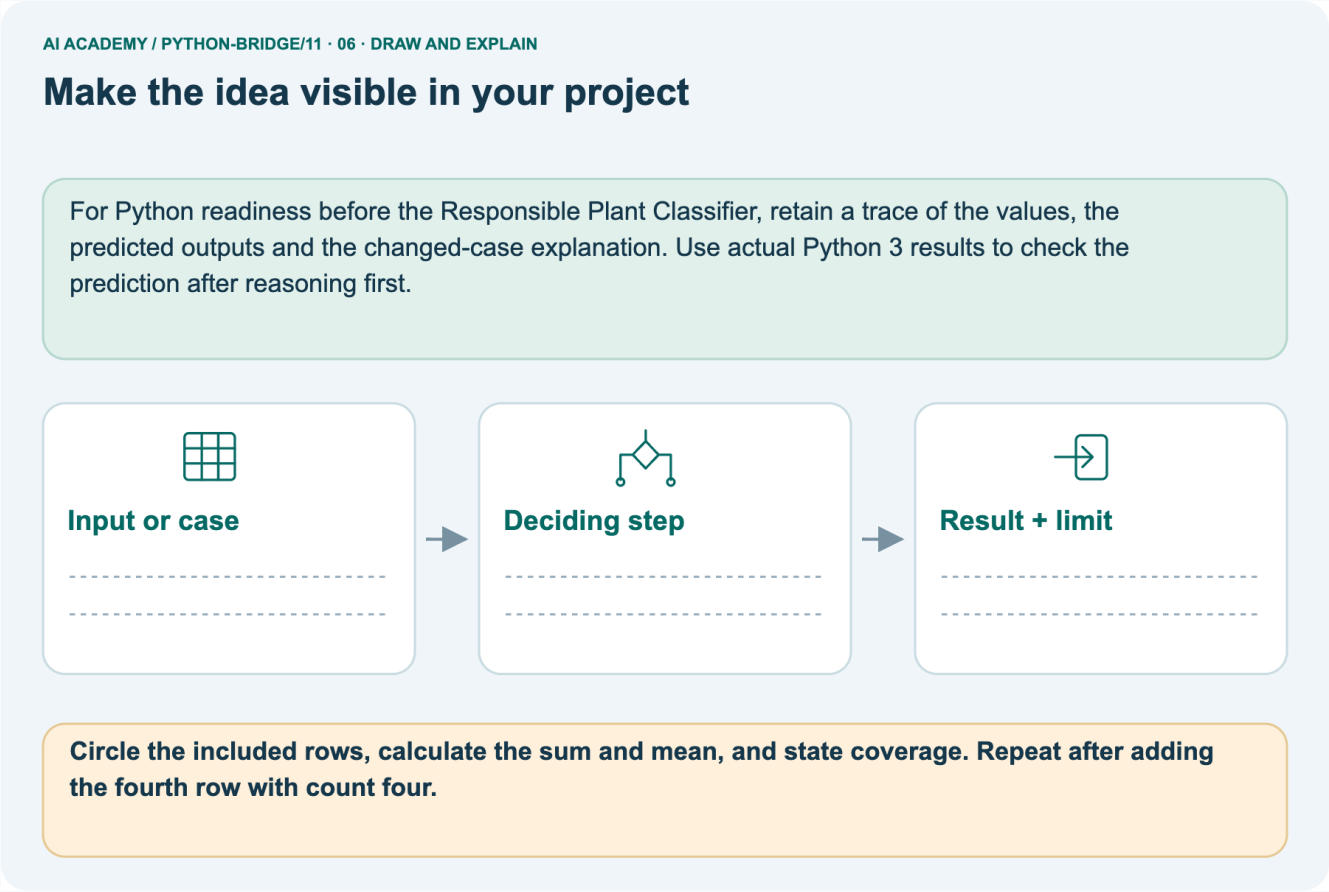
Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Limit: This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.